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CS470 Final Reflection

This reflection describes what I learned in CS 470 and how the course supports my professional goals. In this class, I worked on migrating a full stack web application called the Questions app into the cloud using Amazon Web Services. Through this project, I learned how modern cloud applications are designed, built, and secured. The goal of the course was not just to complete a technical task, but to understand how cloud systems work together in a real-world environment. I learned how cloud-based designs can improve scalability, reduce costs, and strengthen security. A recorded presentation of this project is available at the following link: [Project Two Conference Presentation: Cloud Development](#).

CS 470 helped me better understand how cloud applications are structured and how different services work together. One of my strengths as a developer is my ability to break down complex systems into smaller parts and explain how they interact. This project requires that skill at every step. I learned how the frontend, backend logic, and database can be separated, so each part has a clear role. This separation makes applications easier to manage and easier to scale. I also gained experience controlling access between services so that each part of the system can only perform the actions it is intended to perform.

Security was an important focus throughout this course. I learned how to limit access to sensitive parts of the application using cloud-based permission controls. Only approved services are allowed to access data or run backend processes. This approach reduces risk and helps protect application data. Because of this experience, I feel more prepared for entry-level roles related to cloud development, backend development, or full stack development using cloud

services. I now have a clearer understanding of how real-world cloud systems are built and secured.

Planning for growth was one of the most valuable lessons I learned in this course. Cloud applications are designed to grow over time, and those decisions must be considered early. One major benefit of cloud development is automatic scaling. The application can handle more users without manual server changes. This is especially useful for applications with unpredictable traffic. If the Questions app were to grow, the cloud services used in this project would automatically adjust to handle increased demand.

Cost management is another important factor when planning for growth. Cloud services follow a pay-for-use model, meaning costs are based on actual usage rather than fixed server capacity. This can reduce expenses for smaller applications or applications with fluctuating traffic. For systems with constant high usage, other approaches may provide more predictable costs. Understanding these tradeoffs helps developers choose the right architecture for long-term success. Security must also scale as applications grow. Additional protections such as stronger user authentication, request limits, and monitoring can be added as usage increases. Cloud tools make it easier to implement these improvements over time.

Overall, CS 470 provided a strong foundation in cloud-based development. By migrating a full stack application to a cloud environment, I learned how modern applications are built, scaled, and secured. The course helped me connect technical concepts to real-world use and gave me confidence in my ability to work with cloud systems. The skills I developed in this class will support my future work in software development and cloud-based roles.